Customer Journey Map: Instructor and Learner Lifecycles

Project: NeuroClass — Multimodal Attendance & Engagement Agent

Module: `2.Requirement Analysis`

Phase: Customer Journey & Interaction Analysis

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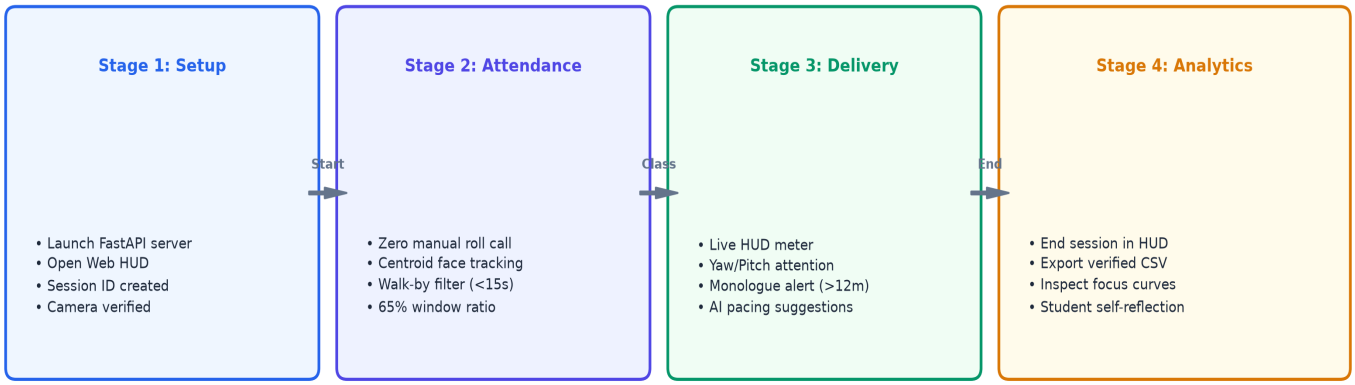
1. Executive Overview

The NeuroClass system transforms the traditional classroom experience by eliminating friction from routine administrative workflows and illuminating student engagement without compromising privacy. This document details the end-to-end journey maps for the two primary actors:

1. **The Course Instructor** (Operational & Pedagogical Flow)
2. **The University Learner** (Participant & Privacy Flow)

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2. Journey Map 1: The Course Instructor



Detailed Interaction Phases (Instructor)

1. Pre-Class Preparation (*t = 0*):

• Instructor opens terminal or launches `python run_from_root.py --agent`.

• The web interface (`http://127.0.0.1:8000`) loads the real-time control room.

• The system checks camera availability and sets up the active session (`SES-YYYYMMDD-HHMMSS`).

2. Lecture Commencement (*t = 0 \text{ to } 15 \text{ mins}*):

• The instructor begins introducing the topic without interrupting to call names.

• The vision module silently detects participants, filters walk-bys (***<15***s), and matches embeddings against `student_db/`.

• Presence counters update in real time on the teacher's HUD.

3. Active Instruction & AI Nudges (*t = 15 \text{ to } 40 \text{ mins}*):

• As the instructor speaks, the acoustic analyzer tracks vocal rhythm and interaction ratios.

• If the instructor speaks continuously for ***>12*** minutes with ***0\%*** student vocal interaction, a gentle toast notification appears:

> *"Teaching Nudge: Continuous monologue detected for 12m. Consider posing an interactive question to reset focus."*

• If class-wide visual attention drops below ***35\%***, the Pedagogical Agent recommends a quick comprehension check.

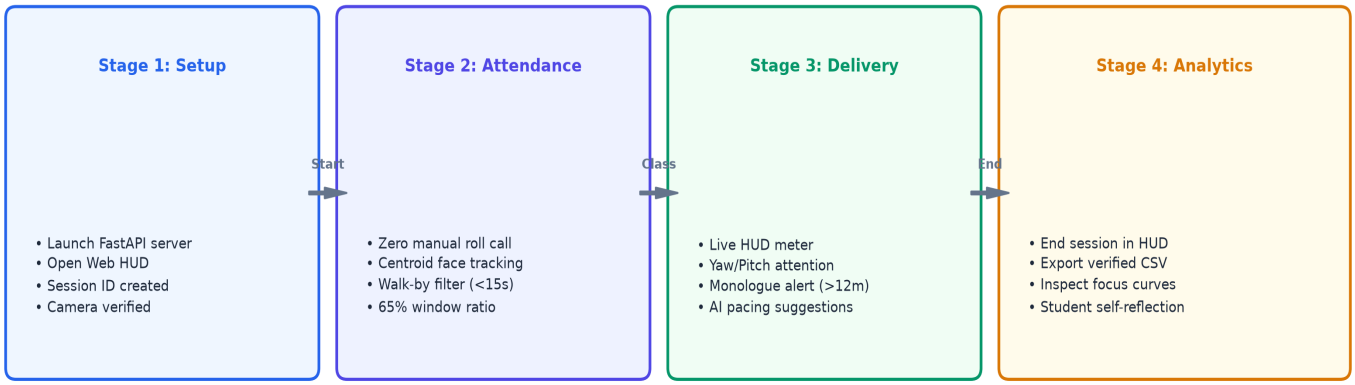
4. Classroom Wrap-Up (*t = 45 \text{ to } 50 \text{ mins}*):

• The instructor clicks **End Session**.

• One-click export generates a tamper-evident CSV with student IDs, names, sustained presence minutes, attendance status (Present, Late, Absent), and average focus score.

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3. Journey Map 2: The University Learner



Detailed Interaction Phases (Learner)

1. Classroom Entry:

• The learner walks into the lecture hall normally. No stopping at a door scanner, no tapping cards, and no signing shared clipboards.

• Transient students walking by the doorway are tracked for ***<15***s and discarded automatically without cluttering the roster.

2. Privacy Assurance During Lecture:

• The camera stream displayed on the instructor screen shows the learner's face blurred (privacy mode active).

• The learner's audio contributions are evaluated strictly for sound pressure and vocal activity. **Words, conversations, and personal remarks are never transcribed, stored, or sent off-device.**

3. Hybrid Identity Enrollment (When Needed):

• If the student is new, the instructor can link their active track (`TRK-001`) to their official ID (`STU-109`) with a single click in the UI (`[IDENTIFY / LINK] Identify / Link`), creating an instant biometric record in `student_db/` for all future classes.

4. Post-Lecture Confidence:

• The student has complete peace of mind that their attendance was registered reliably based on their 40-minute presence, preventing accidental absences caused by clerical errors.